# Zach Zukowski

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## DIGITAL ASSET PRODUCT · TOKENIZED MONEY AND PAYMENTS · INSTITUTIONAL DILIGENCE

· Digital assets since 2016 (10 years) · 1,000+ deal funnel · 100 portfolio companies advised 1-on-1 (50+ multi-session) · LayerZero seed (~80x)
· Thorchain $200M insolvency modeled 13 months ahead · 9 SSRN papers · 25 federal comment letters
· 42 USPTO patent applications across 32 families (Canonical-State Architecture for multi-agent AI research)

## SUMMARY

Ten years in digital assets: independent protocol research from Jul 2016, institutional practice from Dec 2019. Digital-asset product strategist across stablecoins, deposit tokens, tokenized money-market funds, and on-chain payment rails, grounding product calls in control-layer and operator-accountability research. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: payment stablecoins, deposit tokens, tokenized money-market funds, on-chain payment rails · token-economic and scenario modeling, build-versus-partner framework design · OCC, FDIC, FinCEN, Treasury GENIUS, CLARITY, MiCA.

## EXPERIENCE

**Founder** · Tokenization Systems *Feb 2026 to Present*

- Authored a nine-paper SSRN-live research program covering the Control Layer Intensity Index (CLII) for stablecoin gateway and payment-rail concentration, the Minimum Viable Equivalence Pack (MVEP) for institutional tokenization diligence, programmable monetary-policy design (Adaptive Tokenomics), governance concentration, and operational risk in token economies. Three additional papers in development (B5 Bankruptcy-Priority Misaligned Incentives, A7 Control Layer Theory, SVB Reserve Concentration); methodology paper in preparation.
- Contributed 25 comment letters across a federal regulatory program (OCC bank stablecoin engagement RIN 1557-AF41 plus supplemental control-layer extension; Treasury GENIUS Act state similarity; FDIC payment-stablecoin reserve standards RIN 3064-AG19 and stablecoin-subsidiary issuance approvals RIN 3064-AG20; FinCEN AML/CFT FINCEN-2026-0034; FinCEN-OFAC PPSI sanctions FINCEN-2026-0100; joint banking AML/CFT 91 FR 18304); Routing the Dollar prepared for the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); Governance Concentration published at Frontiers in Blockchain.
- Active dialogues across U.S. national-security research, major securities exchanges, central securities depositories, global payments messaging infrastructure, tier-1 bank tokenization platforms, and SEC-registered transfer agents on stablecoin, deposit-token, tokenized-deposit, tokenized money-market-fund, tokenized-commodity, and tokenized-equity deployment.

**Senior Investment Analyst, Digital Assets** · Borderless Capital (RIA; first hire) *Dec 2019 to Feb 2026*

***Product Strategy & Lifecycle***

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.

***Product Systems & Tooling***

- Built an automated SEC-filing and press-release monitoring pipeline (n8n + Apify) covering 30+ Digital Asset Treasury Stocks (DATs), tracking coins held, cash, debt, and share counts (float, adjusted, fully diluted) per issuer to compute multiple on NAV (mNAV) apples to apples across the sector; analyzed ~75 DATs in 2025 and invested in 18; AI-assisted diligence workflows reducing subscription-document review from ~4 hours to ~1 hour per deal.
- Built apples-to-apples multiple on NAV (mNAV) methodology for Digital Asset Treasury Stocks (DATs) and shipped a daily mNAV dashboard sourcing SEC filings and press releases; standardized treatment of non-crypto assets in the numerator across issuers (issuer-reported mNAV figures use inconsistent conventions that undermine cross-DAT comparison).

***Risk & Exception Management***

- Predicted Thorchain's **$200M insolvency 13 months in advance**, leading to a no-exposure posture before the predicted failure mode materialized.
- Prevented hostile governance capture on a top-25 blockchain via voting-path simulation under coalition-attack scenarios; surfaced the vulnerability and mitigation route before exploitation.
- Identified ~15% misreporting of circulating supply in a top-25 Layer 1 protocol by independently modeling its tokenomics (emission schedule, vesting, burn dynamics); the discrepancy had propagated to CoinMarketCap and CoinGecko, and was corrected after partnering with the protocol's foundation engineers.

***Market Discovery & Sourcing***

- Sourced the **LayerZero seed round** investment, returning ~80x; the underlying cross-chain interoperability thesis preceded the firm's Cross-Chain Ecosystem Fund by two years.
- Shipped 250 portfolio investments (~25% deployment rate) across ~10 institutional funds (~$500M AUM) from a 1,000+ deal pre-seed-to-Series-C funnel, with build/partner/hold recommendations to committee.

**Blockchain Research Analyst (Independent)** · Self-Employed *Jul 2016 to Present*

- Independent digital-asset research since Jul 2016, three years before joining Borderless Capital: reviewed 250+ protocol whitepapers, diligenced 1,000+ project and token-offering pitches, consulted 150+ projects on protocol design and tokenomics, and invested in 50+ token offerings.
- Hands-on validation across 30+ blockchains under live conditions: liquidity, fees, settlement finality, bridging, governance execution, and custody and security trade-offs. 30+ conferences and bank summits across the U.S., Europe, Asia, and Dubai.

**Analyst Intern** · Greenway Solutions (fraud-control consulting; client: USAA) *May to Sep 2016*

- Benchmarked fraud controls and the security-versus-friction tradeoff across online, mobile, and call-center channels at nine leading U.S. banks: authentication and password-reset controls, account-opening data requirements, and funds-availability and clearing timing; built the scoring framework and delivered the executive summary.

## SELECTED RESEARCH

- **Routing the Dollar.** Prepared for Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways with Tier 1 a 56% / 44% Coinbase-Circle duopoly; on Ethereum, transfer volume doubled while unique counterparties fell 25% and cross-gateway connections dropped from 5.8% to 2.6%.
- **Who Controls Agentic Payments?** First control-layer measurement of agentic payments (x402). May 2026 facilitator concentration: Base HHI 6,009 by count and 4,040 by dollar volume; Solana 4,203 and 4,915, all above the 2,500 highly-concentrated threshold at under 14 months of market age.
- **FSB AI Sound Practices response.** Filed 2026-07-01; the program's first international standard-setter filing. Three practice additions (composition-chain documentation, execution-authority evidence, dependency-concentration monitoring) plus the invited nonbank case study on settlement facilitators.
- **Three Tokenization Strategies.** Each of three strategies satisfies different MVEP categories with measurably different strength (Regulated Collateral Circuit, Multi-Issuer Yield Stack, Tokenized Equity Rails); Securitize as cross-strategy concentration risk.
- **Seven Dollars.** Cross-product framework for major dollar-denominated digital products and how dollars move across them; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).
- **Dollar v3.** Four digital-dollar objects (payment stablecoins, tokenized deposits, deposit tokens, yield wrappers); typology for inheritance of control environments, backstop expectations, and failure modes; load-bearing for PM scoping of product shape and counterparty topology.
- **Minimum Viable Equivalence Pack.** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at $40B scale. Editor-invited submission to the Journal of Financial Regulation.
- **Adaptive Tokenomics.** Industrial control engineering (PID, Proportional-Integral-Derivative) applied to token-supply emission as programmable monetary-policy mechanism; 806-run agent-based simulation battery; PID emission control cuts worst-case node-count variance from CV (coefficient of variation) 0.544 (static emission) to 0.087 (PID), functioning as downside insurance under competitor-shock attack.
- **Operational Risk in Token Economies.** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 36 control objectives) mapped to a 624-configuration battery; Lido V3 staking-vault case study with custody and reporting controls relevant to tokenized payment platforms.
- **Auditing Governance Concentration Beyond Token Allocation.** Published at Frontiers in Blockchain (2026). Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's d = 0.65, a directionally robust medium effect).
- **Who Burns the Tokens?** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.

## TECHNICAL

**Methods:** policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design
**Data:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring
**Stack:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino, BigQuery), JavaScript / TypeScript; 15 MCP servers and 27 API services wrapping the data sources above plus GitHub; Claude.ai + Claude Code + OpenAI Codex + Cursor with 78-skill library and 33-file canonical-state architecture; Jev (TypeSafe) typed-judgment offload; local models via Ollama (IBM Granite, Qwen)

## RECOGNITION

- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- Introduction to Tokenomics article publicly endorsed by Binance founder CZ on day of publication

## FRAMEWORKS AUTHORED

- Control Layer Intensity Index (CLII), Minimum Viable Equivalence Pack (MVEP), MVEP-gated Conditional Pass-Through (MGCPT; AG19 Rec 7), Three-Category Reserve Disclosure (3CRD).

## EDUCATION

**High Point University** | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President · Lingnan University, Hong Kong (exchange) *2018*